YANLIN LI

School of Pharmaceutical Sciences, Chongqing University, Chongqing, China
liyanlin@cqu.edu.cn

Education

Chongqing University **Sept. 2023 - Present**

- Master of Pharmaceutical Sciences (Major in Bioinformatics and Drug Design)
- GPA: 3.87/4.0 (Rank: 2/102)

Chongqing University **Sept. 2019 - Jun. 2023**

- B.S. of Pharmaceutical Science
- GPA: 3.72/4.0 (Rank: 2/47)

Research Experience

Training a generative peptide model based on diffusion

As the primary investigator **Feb. 2025 – Present**

- Developed a diffusion-based generative model for peptide design, extending the PepGLAD framework.
- Mainly curated an advanced peptide-target complex dataset, significantly expanding the number of peptide-protein complexes used for training. Fine-tuned a generative model on diverse datasets, achieving a higher success rate.
- Integrated an external protein property prediction model as a constraint, guiding the generation toward peptides with higher therapeutic relevance.

Updating an online database of synthetic binding protein (SYNBIP 2.0)

As the primary investigator **Jul. 2023 - Sept. 2024**

- Updated the SYNBIP database to tackle key issues in synthetic binding protein development, including fragmented data, lack of standardization, absence of epitope information, and limited scaffold diversity that restricts efficient design.
- To address the above challenges, we employed a combination of manual literature curation, protein-protein docking, structural alignment, artificial intelligence, and automated data mining through web crawling.
- Access address of SYNBIP: https://synbip.idrblab.net/.

Exploring and analyzing the sequence space of synthetic binding proteins using deep learning models

As the primary investigator **Nov. 2022 - Mar. 2023**

- Applied an advanced deep learning model to design proteins based on SYNBIP database, addressing the limitations of site-specific mutations in traditional protein engineering. The designed proteins showed enhanced sequence diversity, solubility, stability, and binding affinity.

Company-sponsored Research Training

China Resources Double Crane Pharmaceutical Company Limited

Trained in the Pharmacology Group **Sept. 2024 – Present**

- Evaluated and deployed AI models for pharmacology-related tasks, providing technical support for early drug discovery pipelines.

- Performed early-stage drug screening via cellular assays, mouse toxicity and efficacy studies. conducted statistical analysis to support clinical trial applications.

Publications

- **Li Y**, Li F, Duan Z, Liu R, Jiao W, Wu H, Zhu F, Xue W. SYNBIP 2.0: epitopes mapping, sequence expansion and scaffolds discovery for synthetic binding protein innovation. *Nucleic Acids Res.* **2025**;53(D1):D595-D603.
- **Li Y**, Duan Z, Li Z, Xue W. Data and AI-driven synthetic binding protein discovery. *Trends Pharmacol Sci*. **2025**;46(2):132-144.
- **Li Y**, Jiao W, Liu R, Deng X, Zhu F, Xue W. Expanding the sequence spaces of synthetic binding protein using deep learning-based framework ProteinMPNN. *Front Comput Sci*. **2025**;19: 195903.
- Mijit A, **Li Y**, Xue W. Predicting Epitopes of Synthetic Binding Proteins Using Protein Structure Prediction and Protein-Protein Docking. *Methods in molecular biology (Clifton, NJ)*. **2025**, 2937, 245-259.
- Mijit A, Wang X, **Li Y**, Xu H, Chen Y, Xue W. Mapping synthetic binding proteins epitopes on diverse protein targets by protein structure prediction and protein-protein docking. *Comput Biol Med*. **2023**;163:107183.
- Wang X, Li F, Qiu W, Xu B, **Li Y**, Lian X, Yu H, Zhang Z, Wang J, Li Z, Xue W, Zhu F. SYNBIP: synthetic binding proteins for research, diagnosis and therapy. *Nucleic Acids Res.* **2022**;50(D1):D560-D570.

Competitions

- The 19th "Challenge Cup" National College Student Academic Science and Technology Competition
  As the project leader **Third Prize at the Provincial Level**
- Chongqing University Student Innovation Competition (2025)
  As the project leader **Second Prize at the School Level**
- National College Student Social Practice and Science & Technology Competition on Energy Conservation and Emission Reduction (2022) **Second Prize at the National Level**
- National Undergraduate Mathematics Competition (2020)
  Independent competition **Second Prize at the Provincial Level**

Honors & Awards

- The 8th Top 10 Academic Stars of Chongqing University.
- First-Class Postgraduate Scholarship for the 2022-2023 academic year.
- Excellent League Member of Chongqing University (every academic years from 2019 to 2024).
- Excellent Student Cadres of Chongqing University for the 2019-2020 and 2023-2024 academic year.
- National Scholarships for Undergraduate Students for the 2021-2022 academic year (Top 0.2% in China).
- Language proficiency (IELTS = 6).

Professional Competence

- Programming: Python, Shell, Anaconda.
- Computational biology skills: protein molecular dynamics simulation, molecular docking, protein-protein interaction analysis.
- Computational biology softwares: Amber (Linux version), Rosetta, AutoDock, Jalview, Chimera, PyMOL.

- Experimental Skills: Oral gavage in mice; cell culture and cell-based assays (including compound toxicity testing)
- Research Tools: EndNote, Origin, Gephi, GraphPad Prism.
- Good experience in writing academic reports and publications.